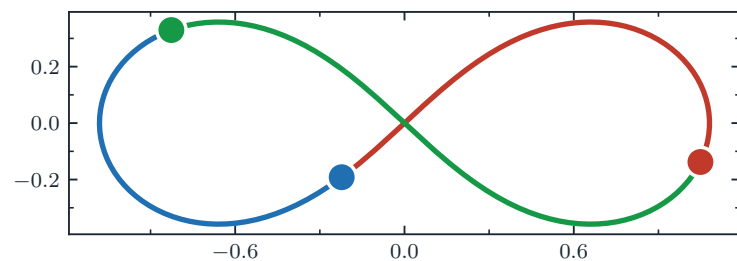
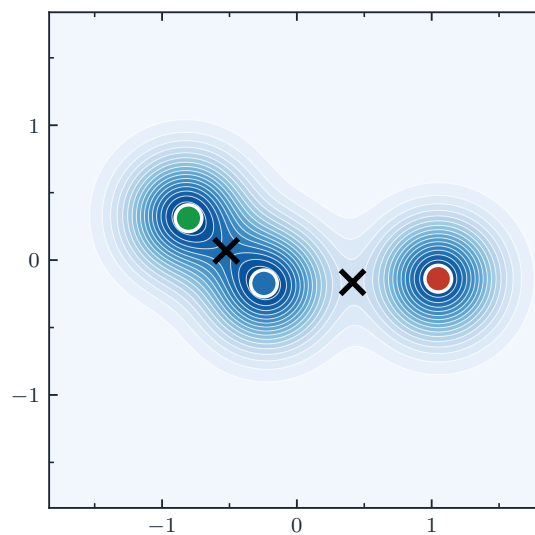


Vineyard monodromy of order 3 in the three-body problem, and the knot it closes to

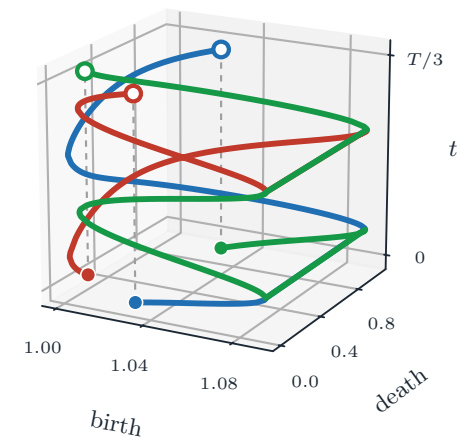
(a) the figure-eight choreography:
three bodies, one curve, over $T/3$



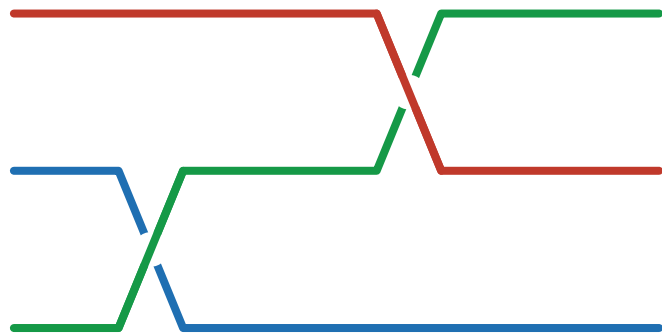
(b) the mass density $\rho(x; t)$: its three
maxima ($\bullet$) and the two merge saddles ($\times$)



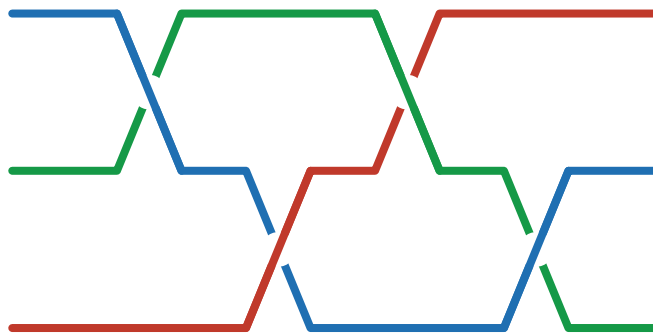
(c) the closed vineyard: each vine ($\bullet$) ends
on a *different* start point ($\circ$)



(d) braid of the *trajectories*: $\sigma_1\sigma_2^{-1}$



(e) braid of the *vineyard*: $(\sigma_2^{-1}\sigma_1)^2$, conjugate to the square of (d)



(f) the same braid, closed up: this is the
figure-eight knot 4_1 , $\Delta(t) = t - 3 + t^{-1}$

